UNIVERSITY OF RAJSHAHI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CSE4261 NEURAL NETWORKS AND DEEP LEARNING

Assignment-10

Author:
Name: MD. MIJU AHMED
ID: 2010676104

Supervisor:
SANGEETA BISWAS
ASSISTANT PROFESSOR

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Abstract

This report investigates the fundamental differences in the generative capabilities of three autoencoder architectures. By training these models and then feeding their decoders random noise, we seek to answer the following questions:

1. What kind of images are generated by the decoder of a **standard Autoencoder** when passed noise vectors drawn from a normal distribution with a mean of 5?

2. How do the generated images from a **Denoising Autoencoder's** decoder differ when given the same random noise?

3. After training and evaluating a **Variational Autoencoder (VAE)**, how does it structure its latent space differently from the other models?

4. Can a VAE's decoder generate meaningful new images when passed noise vectors from the same distribution, and why does its performance differ from the standard and denoising autoencoders?

The findings from these experiments, supported by an analysis of each model's loss function, highlight the critical role of latent space regularization in transforming a reconstructive model into a true generative model.

1 Introduction

Autoencoders are a powerful class of unsupervised neural networks, primarily used for dimensionality reduction and feature learning. They consist of an encoder that compresses input data into a low-dimensional latent space and a decoder that reconstructs the original data from this latent representation. While all autoencoders share this basic structure, their internal workings and objectives differ significantly, which profoundly impacts their ability to function as generative models.

This report compares three types of autoencoders: the standard Autoencoder, the Denoising Autoencoder, and the Variational Autoencoder (VAE). The central experiment involves testing their decoders' ability to generate novel images from random noise vectors drawn from a normal distribution with a mean of 5 and a standard deviation of 1. This investigation will reveal which architectures are merely capable of reconstruction and which possess true generative power.

2 Generation Images from My Custom Autoencoder

2.1 Concept and Loss Function

A standard autoencoder is trained to minimize reconstruction error. Its sole objective is to learn an efficient compression (encoding) and decompression (decoding) of the training data. The model does not impose any structure on how the latent space is organized. The loss function is typically Mean Squared Error (MSE):

$$L = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \tag{1}$$

where x is the original input and $\hat{x}$ is the reconstructed output. Because its only goal is to minimize this pixel-wise error, the decoder only learns to map specific, known latent points back to valid images, leaving large "gaps" in the latent space.

2.2 Results

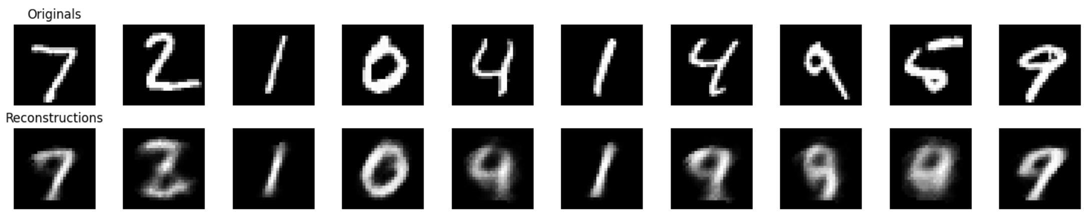


Figure 1: Some images reconstructed by the decoder of a standard autoencoder from random noise vectors.

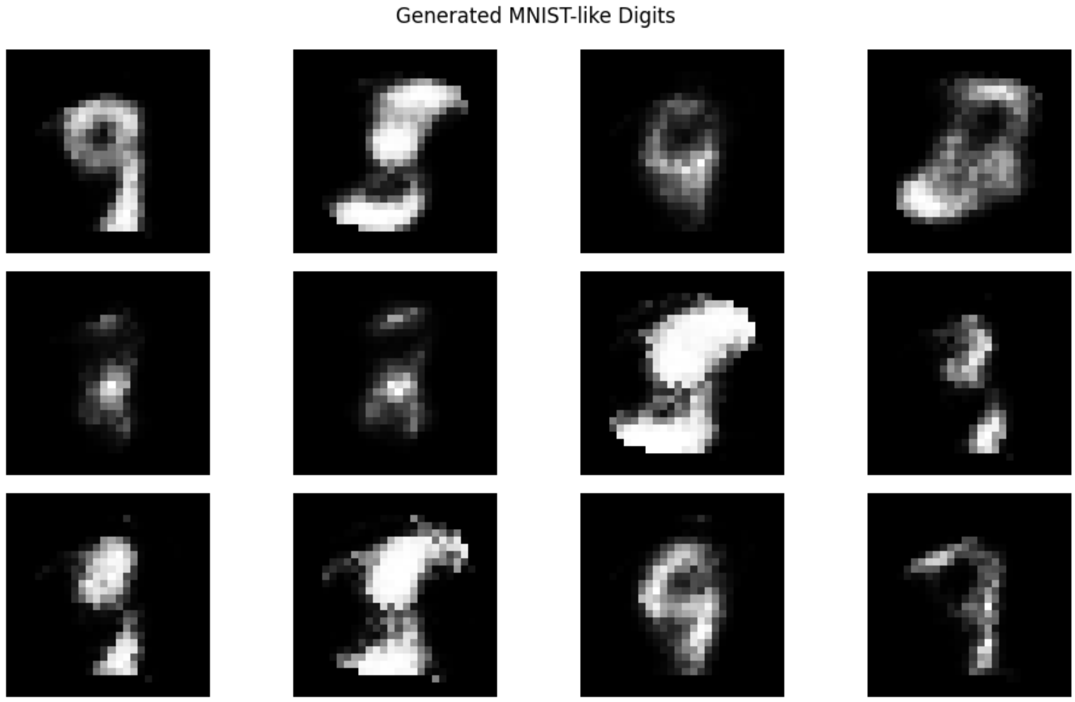


Figure 2: Some images generated by the decoder of a standard autoencoder from random noise vectors.

Findings: As predicted, the images generated by the standard autoencoder's decoder (Figure 1) are unrecognizable noise. Feeding the decoder a random vector that falls into an empty region of the latent space results in a meaningless output. This confirms that the standard autoencoder is a **reconstructive, not generative**, model.

3 Generation from a Denoising Autoencoder

3.1 Concept and Loss Function

A denoising autoencoder is trained to reconstruct a clean image from a corrupted (noisy) input. This forces it to learn more robust features. The loss function is identical to the standard autoencoder's, but the context is different. The model minimizes:

$$L = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \quad \text{where} \quad \hat{x} = \text{Decoder}(\text{Encoder}(x_{\text{noisy}})) \tag{2}$$

The loss is calculated against the original clean image x, even though the input was a noisy version x_{noisy}. Like the standard autoencoder, it has no mechanism to organize the latent space, so its generative capabilities are expected to be poor.

3.2 Results

Findings: The results in Figure **??** are consistent with our analysis. The generated images are again meaningless noise. While the features learned are more robust for reconstruction, the structure of its latent space is still not suitable for generation. This further confirms that it is also a **reconstructive** model.

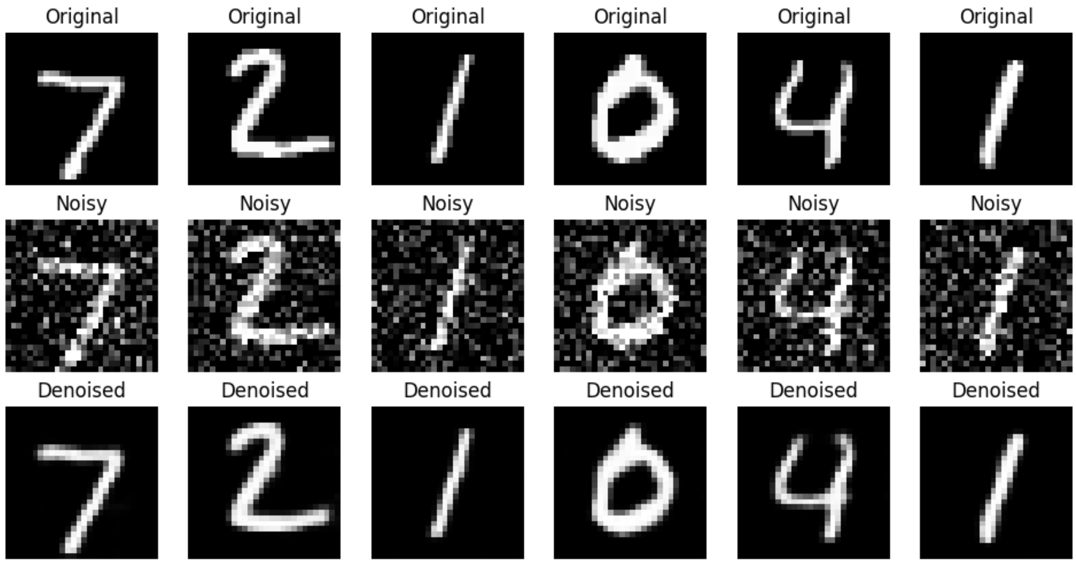
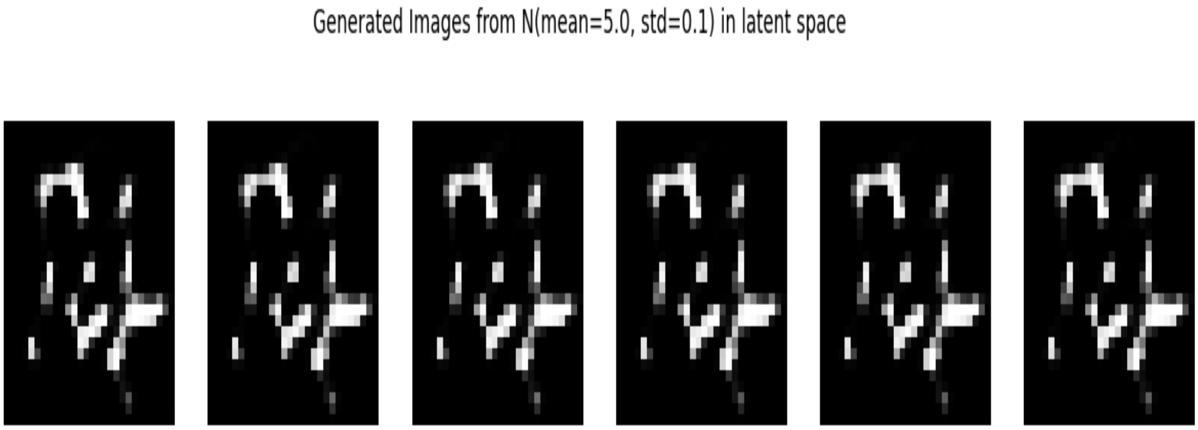
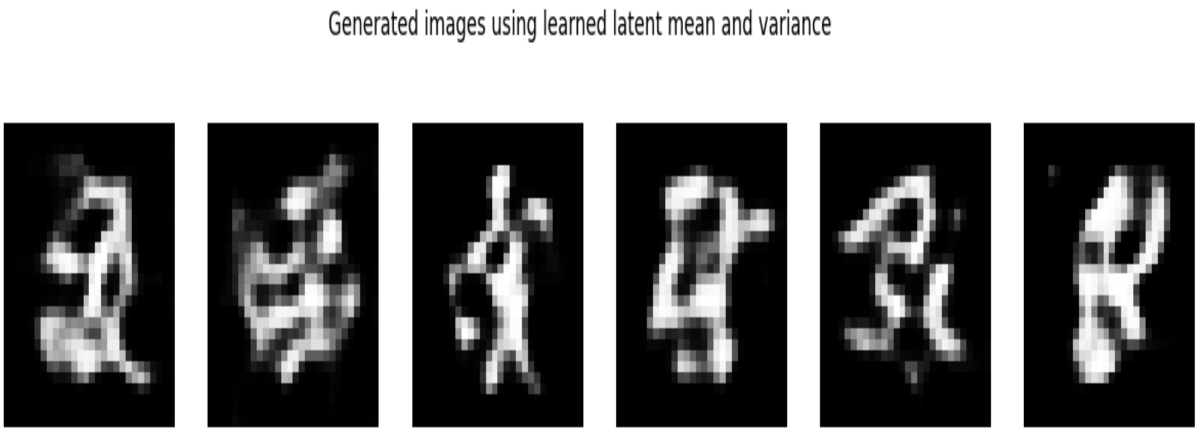


Figure 3: Some images are reconstructed by the decoder of a denoising autoencoder from random noise vectors.



(a) Using mean 5



(b) Using calculated mean

Figure 4: Image generation using Denoising Auto-Encoder

Findings: The results in Figure 4 are consistent with our analysis. The generated images are again meaningless noise. While the features learned are more robust for reconstruction, the structure of its latent space is still not suitable for generation. The generated images' resolution are so poor.

4 Training and Evaluating a Variational Autoencoder (VAE)

4.1 Concept and Loss Function

The Variational Autoencoder is fundamentally different. It is a true generative model. The key innovation is its unique loss function, which has two components: a reconstruction term and a regularization term.

$$L_{\text{VAE}} = L_{\text{reconstruction}} + L_{\text{KL}} \tag{3}$$

- **Reconstruction Loss ($L_{\text{reconstruction}}$):** This is the same as in other autoencoders (e.g., MSE or binary cross-entropy) and ensures the decoded images resemble the originals.
- **Kullback-Leibler (KL) Divergence Loss (L_{KL}):** This is the critical regularization term. The VAE's encoder outputs the parameters of a probability distribution (the mean μ and log-variance $\log(\sigma^2)$). The KL loss forces this learned distribution to be close to a standard normal distribution (mean=0, variance=1). Its formula is:

$$L_{\text{KL}} = -0.5 \sum_{j=1}^{\text{dim}} (1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2) \tag{4}$$

  This term regularizes the latent space, forcing it to be continuous and centered, thereby eliminating the "gaps" found in other autoencoders.

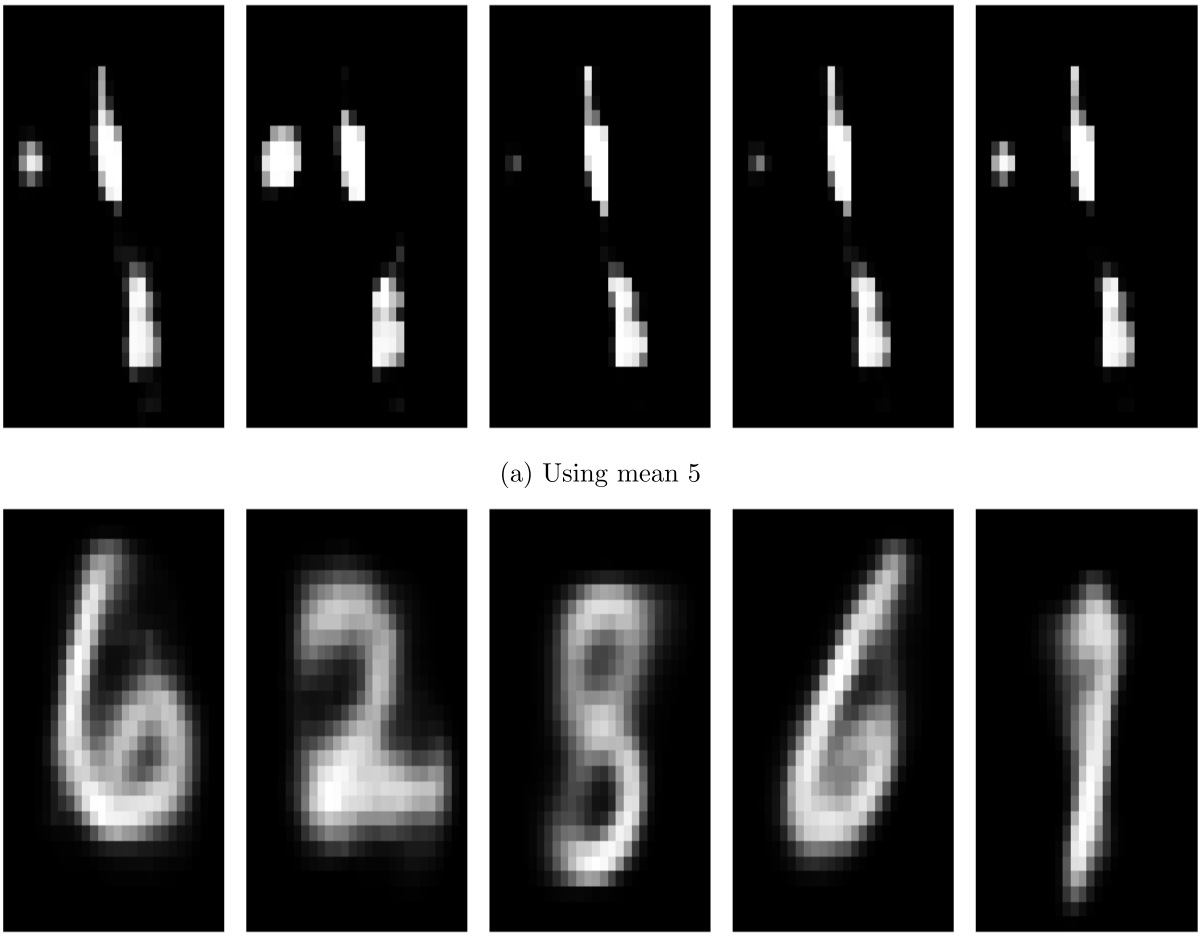
This dual-objective loss function is what enables the VAE to generate new data.

5 Generation from a Variational Autoencoder

5.1 Concept and Analysis

Because the VAE's latent space is regularized by the KL divergence loss to resemble a continuous normal distribution, we can now confidently sample random vectors from that same distribution and expect the decoder to produce meaningful results. The decoder has learned a smooth mapping from this continuous space back to the space of realistic images.

5.2 Results



(a) Using mean 5

(b) Using mean 0

Figure 5: Image generation using Variational Auto-Encoder

Findings: As seen in Figure 5, the VAE successfully generates novel, coherent images that resemble the training data. The images are not perfect copies of any single training example but are new instances that capture the essence of the dataset's classes. This result starkly contrasts with the outputs from the standard and denoising autoencoders and confirms that the VAE is a true **generative** model.

6 Conclusion

This report demonstrated the critical differences between standard, denoising, and variational autoencoders, particularly concerning their generative capabilities. The experiments, supported by an analysis of each model's loss function, showed that standard and denoising autoencoders fail to generate coherent images from random noise because their latent spaces are unstructured.

In contrast, the Variational Autoencoder, through its unique dual-component loss function featuring the KL divergence, learns a continuous and well-organized latent space. This allows its decoder to function as a powerful generative tool, capable of producing novel and plausible images from random input vectors. This investigation clearly illustrates that for a model to be generative, its loss function must not only enforce reconstruction accuracy but also impose a regular, continuous structure on its latent space.

Assignment Github Link : Assignment